10. Give the answer in scientific notation:
A human heart beats at an average of 80 beats per minute. How many times does it beat in i) an hour? ii) a day? iii) a year? iv) 100 years?

Challenging Problems

11. In a map, if 1 inch refers to 120 km, then find the distance between two cities B and C which are $4\frac{1}{6}$ inches and $3\frac{1}{3}$ inches from the city A which lies between the cities B and C.

12. Give an example and verify each of the following statements.
    (i) The collection of all non-zero rational numbers is closed under division.
    (ii) Subtraction is not commutative for rational numbers.
    (iii) Division is not associative for rational numbers.
    (iv) Distributive property of multiplication over subtraction is true for rational numbers. That is, $a(b-c)=ab-ac$.
    (v) The mean of two rational numbers is rational and lies between them.

13. If $\frac{1}{4}$ of a *ragi adai* weighs 120 grams, what will be the weight of $\frac{2}{3}$ of the same *ragi adai*?

14. If $p+2q=18$ and $pq=40$, find $\frac{2}{p}+\frac{1}{q}$.

15. Find x if $5\frac{x}{5}\times 3\frac{3}{4}=21$.

16. By how much does $\frac{1}{\left(\frac{10}{11}\right)}$ exceed $\frac{\left(\frac{1}{10}\right)}{11}$?

17. A group of 1536 cadets wanted to have a parade forming a square design. Is it possible? If it is not possible, how many more cadets would be required?

18. Evaluate: $\sqrt{286225}$ and use it to compute $\sqrt{2862.25}+\sqrt{28.6225}$

19. Simplify: $(3.769\times 10^5)+(4.21\times 10^5)$

20. Order the following from the least to the greatest: 16^{25}, 8^{100}, 3^{500}, 4^{400}, 2^{600}

SUMMARY

- A number that can be expressed in the form $\frac{a}{b}$ where a and b are integers and $b\neq 0$ is called a rational number.
- All natural numbers, whole numbers, integers and fractions are rational numbers.
- Every rational number can be represented on a number line.
- 0 is neither a positive nor a negative rational number.
- A rational number $\frac{a}{b}$ is said to be in the standard form if its denominator b is a positive integer and HCF $(a,b)=1$

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